7.5

Example :

$$I_1 = \int \frac{1}{(x-1)\cdot(x+1)} \cdot dx$$

$$I_2 = \int \frac{1}{x-1} \cdot dx \quad ; \quad I_3 = \int \frac{1}{x+1} dx$$

Note :

$$\int \frac{1}{u} du = \ln|u| + C$$

$$\int \frac{1}{ax+b} = \frac{1}{a} \cdot \ln|ax+b| + C$$

$$\Rightarrow I_2 = \ln|x-1| + C$$

$$I_3 = \ln|x+1| + C$$

$$\frac{1}{(x-1)(x+1)} = \frac{(x+1)-(x-1)}{(x-1)(x+1)} \cdot \frac{1}{2}$$

$$= \left[\frac{(x+1)}{(x-1)(x+1)} - \frac{(x-1)}{(x-1)(x+1)} \right] \cdot \frac{1}{2}$$

$$= \left[\frac{1}{x-1} - \frac{1}{x+1} \right] \cdot \frac{1}{2}$$

$$\Rightarrow \int \frac{1}{(x-1)(x+1)} \, dx = \int \left(\frac{1}{x-1} - \frac{1}{x+1} \right) \frac{1}{2} \, dx$$

$$= \frac{1}{2} \left[\int \frac{1}{x-1} \, dx - \int \frac{1}{x+1} \, dx \right]$$

$$= \frac{1}{2} \ln|x-1| - \frac{1}{2} \ln|x+1| + C$$

$$\frac{1}{(x-1)(x+1)} = \frac{1/2}{x-1} - \frac{1/2}{x+1}$$

partial fraction decomposition

partial fraction

partial fraction

$$\frac{1}{(2x-1)(4x+2)} = \frac{?}{2x-1} + \frac{?}{4x+2}$$

Another way to decompose $\dfrac{1}{(x-1)(x+1)}$

Step 1: Set up the unknowns

$$\frac{1}{(x-1)(x+1)} = \frac{A}{(x-1)} + \frac{B}{(x+1)}$$

Step 2: Solve for the unknowns

$$\Leftrightarrow \frac{1}{(x-1)(x+1)} = \frac{A(x+1) + B(x-1)}{(x-1)(x+1)}$$

$$\Leftrightarrow \frac{1}{(x-1)(x+1)} = \frac{Ax + A + Bx - B}{(x-1)(x+1)}$$

$$\Leftrightarrow 1 = \underbrace{(A+B)}x + \underbrace{(A-B)}$$

$$\begin{cases} A+B = 0 \\ A-B = 1 \end{cases} \Leftrightarrow \begin{cases} B = -A \\ A-(-A) = 1 \end{cases} \Leftrightarrow \begin{cases} A = 1/2 \\ B = -1/2 \end{cases}$$

$$\Rightarrow \frac{1}{(x-1)(x+1)} = \frac{1/2}{(x-1)} + \frac{(-1/2)}{(x+1)}$$

Example: $$\int \frac{1}{(2x-1)(4x+2)} \cdot dx$$

$$\frac{1}{(2x-1)(4x+2)} = \frac{A}{2x-1} + \frac{B}{4x+2}$$

$$\frac{1}{(2x-1)(4x+2)} = \frac{A(4x+2) + B(2x-1)}{(2x-1)\cdot(4x+2)}$$

$$\Leftrightarrow \quad 1 = A(4x+2) + B(2x-1)$$

$$1 = (4A+2B)x + 2A - B$$

$$\begin{cases} 4A + 2B = 0 \\ 2A - B = 1 \end{cases} \Leftrightarrow \begin{cases} A = 1/4 \\ B = -1/2 \end{cases}$$

you can use calculator

$$\int \frac{1}{(2x-1)(4x+2)}\,dx = \int \left(\frac{1/4}{2x-1} + \frac{-1/2}{4x+2} \right) dx$$

$$= \frac{1}{4}\int \frac{1}{2x-1}\,dx - \frac{1}{2}\int \frac{1}{4x+2}\,dx$$

$$= \frac{1}{4}\cdot\frac{1}{2}\ln|2x-1| - \frac{1}{2}\cdot\frac{1}{4}\cdot\ln|4x+2| + C$$

$$= \frac{1}{8}\ln|2x-1| - \frac{1}{8}\ln|4x+2| + C$$

Example:

$$\int \frac{1}{(x-1)^2\cdot(x+1)}\,dx$$

$$\frac{1}{(x-1)(x+1)} = \frac{A}{(x-1)} + \frac{B}{(x+1)}$$

$$\frac{1}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{C}{(x-1)^2} + \frac{B}{x+1}$$

LINEAR FACTOR RULE For each factor of the form $(ax+b)^m$, the partial fraction decomposition contains the following sum of m partial fractions:

$$\frac{A_1}{ax+b}+\frac{A_2}{(ax+b)^2}+\cdots+\frac{A_m}{(ax+b)^m}$$

where $A_1, A_2, \ldots, A_m$ are constants to be determined. In the case where $m=1$, only the first term in the sum appears.

Example:

$$\frac{1}{(x+1)^3\cdot(x-1)^2\cdot(x+2)} = \frac{A}{x+1}+\frac{B}{(x+1)^2}+\frac{C}{(x+1)^3}$$

$$+\frac{D}{x-1}+\frac{E}{(x-1)^2}+\frac{F}{(x+2)}$$

$$\frac{1}{x^3\cdot(x+1)} = \frac{A}{x}+\frac{B}{x^2}+\frac{C}{x^3}+\frac{D}{x+1}$$

Problem: $\int \frac{1}{x^2\cdot(x+1)}\,dx$